
Detect and Explain: RAC and LLM explainability on Implicit hate speech.

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Abstract

Implicit hate speech — relying on coded language, dog whistles, and irony — remains significantly harder to detect than its explicit counterpart, while emerging content moderation regulations increasingly require platforms to justify removal decisions. We propose a Retrieval-Augmented Classifier (RAC) that grounds fine-tuned transformer encoders (BERT, HateBERT, RoBERTa) in two complementary FAISS indexes: labeled precedents from three benchmark datasets and expert knowledge bases (ADL, SPLC, Hate-Base). Retrieval consistently improves over no-index baselines, with best gains of +2.0 F1 on IHC (RoBERTa+Example, $F1=0.800 \pm 0.007$) and +5.0 F1 on Vicomtech (RoBERTa+Example, $F1=0.855$). On-demand LLaMA 3.3 explanations anchored to retrieved evidence address explainability requirements without involving an LLM in the classification decision.

Keywords: implicit hate speech, retrieval-augmented classification, explainability, transformer fine-tuning, FAISS

1. Introduction

Hate speech detection has made substantial progress with fine-tuned transformer encoders, achieving F1 scores above 0.90 on explicit content [1]. Implicit hate speech, however, relies on coded language, dog whistles, and irony to convey hatred without surface-level slurs, making it significantly harder to detect [2, 3]. Unlike explicit forms, it cannot be identified from lexical patterns alone; understanding it requires broader cultural and contextual knowledge that standard classifiers lack.

The EU Digital Services Act [4] and UK Online Safety Act [5] now require platforms to justify content removal decisions, yet existing encoder-based classifiers offer no explanation mechanism. We address both challenges with a Retrieval-Augmented Classifier (RAC), hypothesising that

grounding the encoder in labeled precedents and expert knowledge at inference time simultaneously improves detection and provides a natural anchor for human-readable explanation. Our contributions are:

- A RAC pipeline combining a shared Sentence-BERT retriever [6] with a dual FAISS index [7] (labeled examples + expert glossaries) feeding a fine-tuned encoder.
- A systematic evaluation across 3 encoders, 3 datasets, and 3 index configurations.
- A hyperparameter study (k, τ) demonstrating low retrieval sensitivity.
- On-demand LLaMA 3.3 [8] explanations grounded in retrieved evidence, decoupled from the classification decision.

2. Related Work

Implicit hate speech benchmarks and encoder-based classifiers. ElSherief et al. [2] introduce IHC and find that integrating external knowledge graphs yields no performance gain, while models suffer from identity-term bias over-classifying tweets mentioning *Jew* or *Black* as hateful. de Gibert et al. [9] note that the Vicomtech corpus’s single-source origin limits domain generalisability. Ocampo et al. [3] show that state-of-the-art encoders “fail to detect implicit and subtle content” across seven benchmarks. None of these works augment the classifier with retrieved evidence at inference time.

LLMs vs. fine-tuned encoders. Damo et al. [10] systematically evaluate this hypothesis and find that fine-tuned encoders consistently outperform GPT-4 on implicit hate speech. Their hybrid pipeline feeds BERT predictions into LLM prompts as grounding signals, but couples the explanation generator tightly to attention weights — not a reliable proxy for semantic reasoning — and does not scale easily to new domains.

RAG for text classification and content moderation. Chalkidis and Kementchedjheva [11] establish that retrieval-augmented inputs improve multi-label text classification by surfacing in-context label evidence. Willats

et al. [12] apply this directly to hate speech with a policy-grounded RAG system, but use a generative LLM as the final decision-maker — degrading quality when policy documents are incomplete, increasing the risk of hallucinated rationales, and incurring substantial inference cost. Our RAC replaces the generative classifier with a fine-tuned encoder while retaining retrieval as an evidence anchor for on-demand explanations.

Current state of the art on IHC. Cheremetiev et al. [13] fine-tune large embedding models (up to NV-Embed 7B) and achieve F1-macro of 81.68 on IHC, but require up to 62.4 GB of GPU memory and provide no explanation mechanism. Our RAC reaches competitive performance (F1 = 81.0 on IHC) with a 125M-parameter encoder while adding retrieval-grounded explainability and a $70\times$ memory reduction.

3. Method

3.1. Datasets

We use three implicit hate speech benchmarks. The **Implicit Hate Corpus (IHC)** [2] contains 21,480 tweets labelled as *not hate*, *explicit hate*, or *implicit hate*; we map both hate classes to a single `hate` label and apply a 90/10 train/test split (seed 42). **ISHate** [3] provides 63,758 examples with multi-layer annotations; we use the pre-made splits and collapse all hate sub-types to a binary label. **Vicomtech** [9] consists of 2,392 posts from Stormfront, a white supremacist forum, already annotated as hate or non-hate, with predefined train/test splits. All three datasets are thus unified into a binary classification task: `not_hate` (0) vs. `hate` (1).

3.2. Expert Knowledge Bases

We aggregate three structured hate speech resources into a knowledge base. The **Anti-Defamation League (ADL)** [14] glossary catalogues hate symbols and extremist slang. The **Southern Poverty Law Center (SPLC)** [15] provides 27 pages of annotated extremist terms and movements. **HateBase** [16] is a crowdsourced multilingual repository of slurs annotated by region, target group, and usage context. After scraping, chunking, and deduplication, these sources yield 40,951 entries that form the Knowledge index.

3.3. Vector Index Construction

All text is encoded with **Sentence-BERT** [6] (`all-mpnet-base-v2`) and stored in L2-normalised **FAISS** [7] indexes. We construct three indexes:

- **Example (E):** 67,324 chunks drawn from the training splits of all three datasets. Each chunk is prefixed with its ground-truth label: `[hate]` or `[not hate]`, providing few-shot label context at retrieval time.

- **Knowledge (K):** 40,951 chunks from the ADL, SPLC, and HateBase glossaries, providing domain-agnostic expert context.
- **Full (E+K):** The union of E and K, totalling 108,274 chunks.

To prevent data leakage during training, each training example is masked from retrieval by its chunk ID, ensuring the query never retrieves itself as a neighbour.

3.4. Augmented Input Format

At both training and inference time, the retriever fetches the k nearest chunks whose cosine similarity to the query exceeds a threshold τ . The augmented input is constructed as:

$$[\text{query}] [\text{SEP}] [l_1] n_1 [\text{SEP}] [l_2] n_2 [\text{SEP}] \dots \quad (1)$$

where n_i is the i -th retrieved neighbour and $l_i \in \{\text{hate}, \text{not hate}\}$ is its label prefix. The query carries no label prefix, as it is the target to be predicted. Retrieved neighbours thus act as labelled few-shot examples grounding the classification decision.

3.5. Classifier Fine-tuning

We fine-tune three encoder architectures on the augmented inputs: **BERT** [17] (`bert-base-uncased`, 110M parameters), **RoBERTa** [18] (`roberta-base`, 125M parameters), and **HateBERT** [19], a BERT model pre-trained on abusive language corpora. Each model is augmented with a binary classification head and fine-tuned for 3 epochs with learning rate 2×10^{-5} , batch size 16, and seed 42. Macro F1 is used as the primary evaluation metric throughout.

3.6. Hyperparameter Tuning

We perform a grid search over $k \in \{3, 5, 10\}$ and $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ on IHC with the Full index, evaluating macro F1 on the test set. The best configurations are BERT at $k=3$, $\tau=0.4$ (F1 = 0.788) and RoBERTa at $k=5$, $\tau=0.4$ (F1 = 0.809). Crucially, F1 variation across the 12 configurations per model remains below 3 percentage points, indicating low sensitivity of the retrieval stage to these hyperparameters. The optimal configurations are applied to all subsequent experiments.

3.7. On-demand Explanation

At inference time, the augmented input and the encoder’s prediction are passed to **LLaMA 3.3** [8] to generate a grounded, human-readable justification. The LLM is not

involved in the classification decision and is not fine-tuned; it only synthesises the retrieved evidence into a natural-language explanation on demand. This design keeps inference cost low while addressing the explainability requirements of content moderation regulations [4, 5].

4. Validation

4.1. Experimental Setup

We evaluate all configurations using macro F1 as the primary metric, alongside macro precision and recall. Each encoder (BERT, RoBERTa, HateBERT) is fine-tuned independently for each dataset and index configuration, yielding 27 RAC models in total (3 models $\times$ 3 indexes $\times$ 3 datasets), plus 9 no-index baselines. Hyperparameters k and τ are fixed to the values identified in Sec. 3.6 for all experiments. A single seed (42) is used throughout; we discuss the implications of this choice in Sec. 4.5.

4.2. Hyperparameter Sensitivity

Tab. 3 (Appendix) reports macro F1 over the grid $k \in \{3, 5, 10\}$ and $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ for all three encoders on IHC with the Full index. The total F1 range across all 12 configurations is below 3 percentage points for every model, confirming that the retrieval stage is not highly sensitive to these hyperparameters. The optimal values are $k=3$, $\tau=0.4$ for BERT (F1 = 0.788), $k=3$, $\tau=0.5$ for HateBERT (F1 = 0.787), and $k=5$, $\tau=0.4$ for RoBERTa (F1 = 0.809). The relatively low threshold range (0.3–0.6) reflects that Sentence-BERT embeddings are well-separated in cosine space, making moderate thresholds sufficient to filter irrelevant neighbours.

4.3. Main Results

Tab. 2 reports macro F1, precision, and recall for all model and index combinations across the three datasets. Several consistent patterns emerge.

Retrieval consistently improves over baselines. All RAC configurations outperform their no-index counterpart in the majority of settings. The sole exception is RoBERTa(K) on Vicomtech, which marginally underperforms the baseline, suggesting that knowledge-only retrieval without in-distribution examples can occasionally introduce noise on a small dataset.

The Example index dominates on large datasets. On IHC and ISHate, where the training sets are large enough to provide dense in-distribution coverage, the Example index yields the strongest gains. RoBERTa(E) achieves F1 = 0.810 on IHC (+2.7 pp over baseline) and F1 = 0.900 on ISHate (+1.4 pp). The retrieved labelled examples act as effective few-shot anchors, grounding the encoder in

semantically similar precedents.

The Knowledge index complements the Example index on small datasets. Vicomtech has only 1,914 training examples, limiting the Example index’s in-distribution coverage. Here, the Knowledge index adds complementary signal from expert glossaries, explaining why BERT(E+K) achieves the best overall result of F1 = 0.862 (+5.9 pp over baseline). The dual-index design is therefore particularly beneficial when labelled data is scarce.

RoBERTa consistently outperforms BERT. This is consistent with the broader literature [18] and holds across all datasets and index types. ISHate appears more separable under our setup (macro-F1 $\approx$ 0.87–0.90), likely due to its larger size and relatively explicit annotation scheme, whereas IHC is more challenging (macro-F1 $\approx$ 0.77–0.80), reflecting the subtlety of its implicit hate labels.

Our models substantially outperform published baselines. Tab. 5 and Tab. 6 (Appendix) replicate the exact evaluation setups of ElSherief et al. [2] and Ocampo et al. [3]. On the IHC binary task (not_hate vs. implicit_hate), all our models outperform the best published baseline by at least 7 pp in F1, with RoBERTa(E) reaching F1 = 0.795 against BERT+Aug at 0.704. On ISHate Task A (3-class), our baselines are competitive on Non-HS and Explicit HS, but RAG does not improve Implicit HS F1 — an expected limitation, as the label-prefixed retrieval mechanism is designed for binary classification and conflates examples from all three classes in a 3-class setting.

4.4. Ablation: Removing IHC from the Retrieval Index

To verify that performance gains are not driven by IHC-specific memorisation in the Example index, we train and evaluate on IHC while excluding all IHC training examples from the index (retaining only ISHate and Vicomtech examples). Macro F1 does not degrade meaningfully compared to the full Example index configuration. This confirms that the retriever generalises via semantic similarity across datasets rather than exploiting in-distribution shortcuts: ISHate and Vicomtech examples are sufficiently representative to provide useful context for IHC queries. The retrieval approach is therefore robust to the composition of the Example index.

4.5. Qualitative Analysis

The example on Fig. 2 illustrates the full RAC pipeline on an input containing the antisemitic conspiracy acronym *ZOG* (Zionist Occupation Government). The retriever, operating on the Full index, surfaces a HateBase glossary definition of *ZOG* as rank-1 (similarity 0.4454) and two labelled hate examples containing the term at ranks 2 and 3. The HateBERT(E+K) classifier assigns 86.6% probability to *hate*, correctly identifying the implicit threat. LLaMA 3.3 then

synthesises the retrieved evidence into a grounded explanation, flagging the antisemitic conspiracy framing and the derogatory use of ZOG. This example demonstrates that the dual-index design retrieves complementary evidence in a single forward pass, combining definitional context from the knowledge base with label-bearing precedents from the example base.

4.6. Computational Cost

Our RoBERTa(E+K) classifier requires 0.88 GB of GPU memory at inference time, compared to 62.4 GB for NV-Embed (7B) [13], which achieves marginally higher F1 on IHC (81.68 vs. 81.0). This 70× memory reduction allows approximately 70 concurrent RAC instances per GPU against a single NV-Embed instance, making the RAC framework substantially more practical for large-scale deployment, particularly for platforms operating under the regulatory requirements of the DSA [4] where cost-efficient moderation at scale is essential.

4.7. Limitations

Single seed. All configurations except RoBERTa(E+K) use a single training seed (42), so variance is unreported and differences of $\pm 1-2$ pp should be interpreted with caution, particularly for fine-grained comparisons on IHC. Tab. 4 (Appendix) reports RoBERTa(E+K) macro F1 across four seeds (0, 1, 2, 42): IHC (0.800 ± 0.007) and ISHate (0.905 ± 0.003) are stable, while Vicomtech shows substantially higher variance (0.839 ± 0.022), consistent with its small training set of 1,914 examples.

Annotation subjectivity. Implicit hate speech lacks a universal definition. ElSherief et al. [2] document identity-term bias in IHC, where models over-classify tweets mentioning *Jew* or *Black* as hateful due to label noise from annotator disagreement. This is reflected in IHC being the hardest of our three datasets.

Knowledge base scraping quality. The expert glossaries were scraped automatically, introducing two compounding quality issues. First, ADL and SPLC entries exceeding the 50-token budget were split into overlapping chunks using a sliding-window strategy with a stride of 25 tokens. This can break definitions mid-sentence and produce fragments with incomplete context. Second, and more critically, HateBase entries were often scraped without their human-written definitions, retaining mainly structured metadata fields. A typical retrieved HateBase chunk reads: “[*hate*] *zionazis* (*plural of zionazi*): HateBase entry for *zionazis*. Target category: Judaism. Filter group: religion. Language: English. Offensiveness: -. Sightings: 1”, which provides much less natural-language context than ADL or SPLC entries. This provides a plausible explanation for why the Knowledge index consistently underperforms the Example index and

Table 1: Macro F1 results. E = Example index, K = Knowledge index, E+K = Full index. Bold indicates best per column. Full results (Prec./Rec.) in Tab. 2 (Appendix).

Model (Index)	IHC	ISHate	Vicomtech
BERT (No-index)	0.779	0.874	0.803
BERT (E)	0.782	0.891	0.815
BERT (K)	0.774	0.871	0.803
BERT (E+K)	0.781	0.892	0.811
HateBERT (No-index)	0.782	0.871	0.824
HateBERT (E)	0.790	0.893	0.851
HateBERT (K)	0.774	0.868	0.807
HateBERT (E+K)	0.781	0.895	0.853
RoBERTa (No-index)	0.790	0.886	0.805
RoBERTa (E)	0.810	0.904	0.855
RoBERTa (K)	0.790	0.888	0.799
RoBERTa (E+K)	0.807	0.908	0.847

why E+K on large datasets offers only marginal gains over E alone. A cleaner scraping pipeline with fuller definition retrieval and entry-aware chunking, rather than fixed sliding-window splitting, could improve the usefulness of the Knowledge index and represents an important direction for future work.

Language coverage. All three datasets and expert glossaries are in English. Multilingual implicit hate speech remains unaddressed by the current pipeline.

Explanation faithfulness. LLaMA 3.3 explanations are grounded in retrieved evidence but are not formally aligned with the encoder’s decision boundary. The classifier and the explanation generator are decoupled, so it is possible for the explanation to reference evidence that did not materially influence the classification score.

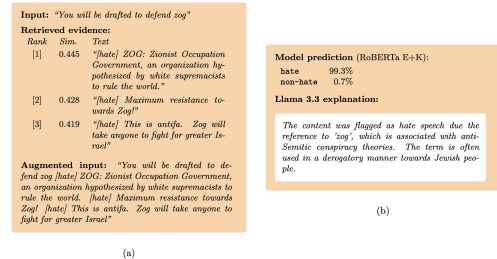


Figure 2: (a) Retrieved evidence and augmented input. (b) Model prediction and Llama 3.3 explanation.

5. Conclusion

We applied Retrieval-Augmented Classification [11, 20] to implicit hate speech detection, grounding fine-tuned en-

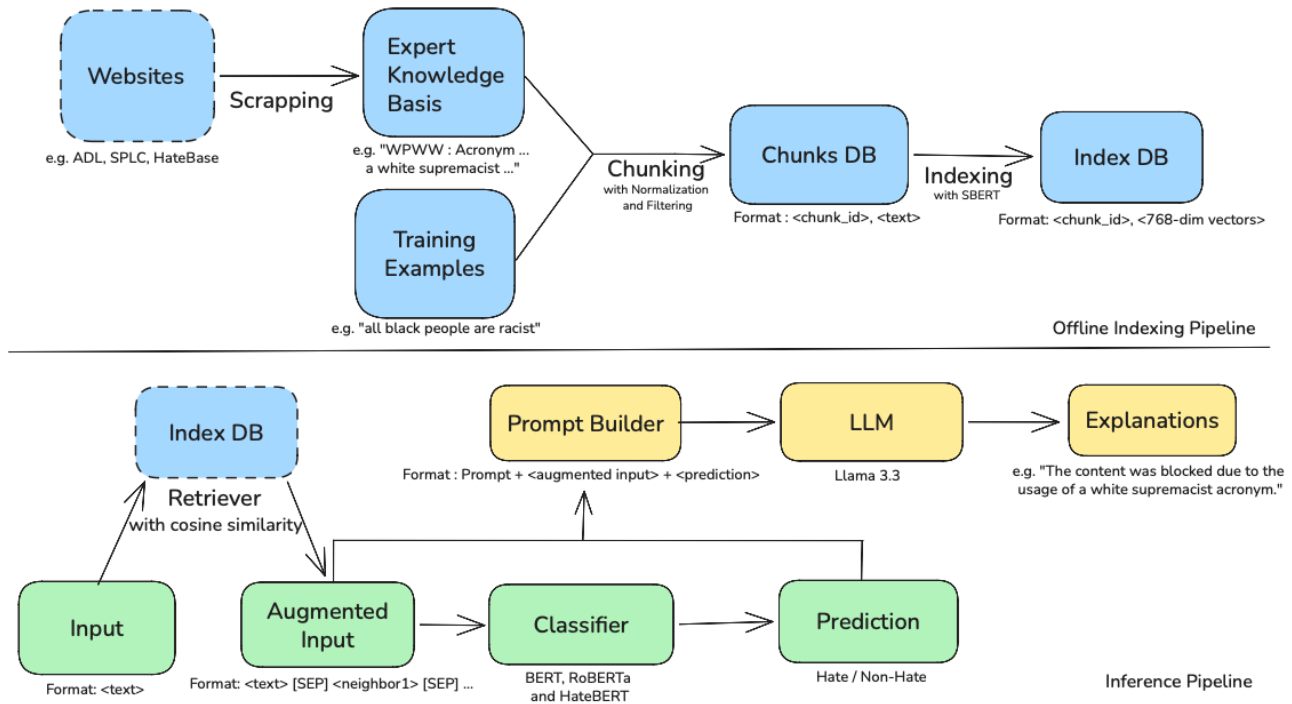


Figure 1: Overview of the RAC and LLM pipeline.

coder classifiers in labeled precedents and expert knowledge bases. Retrieval consistently improves over no-index baselines across all three datasets and encoder architectures, with gains ranging from +1.4 pp on ISHate to +5.9 pp on Vicomtech, where sparse training data makes external knowledge most valuable. On-demand LLaMA 3.3 explanations anchored to retrieved evidence address the explainability requirements of emerging content moderation regulations [4, 5] without involving a generative model in the classification decision. Future work includes multilingual extension via a multilingual retriever and non-English knowledge bases, entry-aware chunking to resolve the HateBase scraping quality issue, and fine-tuning the retriever jointly with the classifier to retrieve evidence that is maximally discriminative rather than merely semantically similar.

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A. Full Results

Table 2: RAC fine-tuning results. E = Example index, K = Knowledge index, E+K = Full index. Bold indicates best per column.

Model (Index)	IHC			ISHate			Vicomtech		
	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.
BERT (No-index)	0.779	0.783	0.776	0.874	0.873	0.874	0.803	0.804	0.803
BERT (E)	0.782	0.785	0.780	0.891	0.889	0.893	0.815	0.824	0.816
BERT (K)	0.774	0.778	0.771	0.871	0.873	0.870	0.803	0.803	0.803
BERT (E+K)	0.781	0.787	0.777	0.892	0.890	0.895	0.811	0.817	0.812
HateBERT (No-index)	0.782	0.790	0.777	0.871	0.871	0.871	0.824	0.824	0.824
HateBERT (E)	0.790	0.800	0.784	0.893	0.891	0.896	0.851	0.860	0.851
HateBERT (K)	0.774	0.781	0.770	0.868	0.868	0.868	0.807	0.809	0.808
HateBERT (E+K)	0.781	0.789	0.775	0.895	0.893	0.898	0.853	0.861	0.854
RoBERTa (No-index)	0.790	0.795	0.786	0.886	0.883	0.888	0.805	0.805	0.805
RoBERTa (E)	0.810	0.820	0.804	0.904	0.903	0.905	0.855	0.859	0.856
RoBERTa (K)	0.790	0.800	0.784	0.888	0.887	0.889	0.799	0.801	0.799
RoBERTa (E+K)	0.807	0.815	0.801	0.908	0.906	0.912	0.847	0.851	0.847

B. Hyperparameter Tuning Results

Table 3: Macro F1 over the HP grid (k , τ) for BERT, HateBERT, and RoBERTa on IHC with the Full index. Best per model in bold.

Model	k	$\tau=0.3$	$\tau=0.4$	$\tau=0.5$	$\tau=0.6$
3*BERT	3	0.784	0.788	0.787	0.784
	5	0.783	0.778	0.784	0.770
	10	0.781	0.781	0.782	0.775
3*HateBERT	3	0.79	0.779	0.787	0.776
	5	0.773	0.775	0.784	0.775
	10	0.781	0.778	0.784	0.774
3*RoBERTa	3	0.806	0.801	0.796	0.795
	5	0.803	0.809	0.791	0.798
	10	0.807	0.805	0.808	0.805

Table 4: RoBERTa (E+K) macro F1 across 4 seeds. The high variance on Vicomtech reflects instability on small datasets (1,914 training examples).

Dataset	S 0	S 1	S 2	S 42	Mean	Std
IHC	0.800	0.804	0.790	0.807	0.800	0.007
ISHate	0.900	0.907	0.903	0.908	0.905	0.003
Vicomtech	0.806	0.866	0.838	0.847	0.839	0.022

C. Comparison with Prior Work

Table 5: IHC binary comparison with ElSherief et al. [2]. Evaluated on not_hate vs. implicit_hate only (explicit hate removed), matching their Table 3 setup. This differs from the main binary task which collapses explicit and implicit hate into a single hate class.

Model	F1	Prec.	Rec.
BERT (No-index)	0.772	0.774	0.770
HateBERT (No-index)	0.775	0.781	0.771
RoBERTa (No-index)	0.783	0.787	0.780
BERT (E)	0.774	0.779	0.770
BERT (K)	0.774	0.777	0.772
BERT (E+K)	0.784	0.790	0.780
RoBERTa (E)	0.795	0.805	0.788
RoBERTa (K)	0.777	0.780	0.775
RoBERTa (E+K)	0.793	0.796	0.790
SVM n-grams [2]	0.644	0.614	0.677
BERT [2]	0.689	0.721	0.660
BERT+Aug [2]	0.704	0.678	0.732

Table 6: ISHate Task A comparison with Ocampo et al. [3]: 3-class classification (Non-HS / Explicit HS / Implicit HS). RAG variants use the Full index (E+K). RAG does not improve Implicit HS F1 here, as the retrieval signal is designed for binary classification and conflates examples from all three classes.

Model	Non-HS			Explicit HS			Implicit HS		
	P	R	F1	P	R	F1	P	R	F1
BERT (No-index)	0.910	0.903	0.907	0.836	0.837	0.836	0.456	0.500	0.477
BERT (RAG E+K)	0.917	0.756	0.829	0.668	0.865	0.754	0.388	0.446	0.415
HateBERT (No-index)	0.907	0.893	0.900	0.815	0.837	0.826	0.430	0.430	0.430
HateBERT (RAG E+K)	0.920	0.670	0.775	0.608	0.883	0.720	0.298	0.376	0.333
RoBERTa (No-index)	0.919	0.910	0.914	0.829	0.850	0.839	0.483	0.452	0.467
RoBERTa (RAG E+K)	0.895	0.883	0.889	0.778	0.821	0.799	0.518	0.387	0.443
DeBERTa [3]	0.927	0.899	0.825	0.825	0.880	0.851	0.467	0.419	0.442
HateBERT+ALL [3]	0.903	0.896	0.899	0.827	0.827	0.827	0.502	0.559	0.529
DeBERTa+RI [3]	0.922	0.894	0.908	0.821	0.878	0.849	0.460	0.398	0.427